21 REVERSIBLE REACTIONS AND EQUILIBRIA

This chapter explores the idea of 'reversibility' in a reaction, and how you can control the conditions of such reactions in order to obtain as much as possible of what you want.

LEARNING OBJECTIVES

- Know that some reactions are reversible and that this is indicated by the symbol ⇌ in equations
- Describe reversible reactions such as the dehydration of hydrated copper(II) sulfate and the effect of heat on ammonium chloride

CHEMISTRY ONLY

- Know that a reversible reaction can reach dynamic equilibrium in a sealed container.
- Know that the characteristics of a reaction at dynamic equilibrium are:
  - the forward and reverse reactions occur at the same rate
  - the concentrations of reactants and products remain constant.
- Understand why a catalyst does not affect the position of equilibrium in a reversible reaction.
- Know the effect of changing either temperature or pressure on the position of equilibrium in a reversible reaction:
  - an increase (or decrease) in temperature shifts the position of equilibrium in the direction of the endothermic (or exothermic) reaction
  - an increase (or decrease) in pressure shifts the position of equilibrium in the direction that produces fewer (or more) moles of gas.

REVERSIBILITY AND DYNAMIC EQUILIBRIA

TWO REVERSIBLE REACTIONS



▲ Figure 21.1 Copper(II) sulfate crystals are split into anhydrous copper(II) sulfate and water on gentle heating.

DEHYDRATION OF COPPER(II) SULFATE CRYSTALS

If you heat blue copper(II) sulfate crystals gently, the blue crystals turn to a white powder and water is driven off. Heating causes the crystals to lose their *water of crystallisation* and white *anhydrous* copper(II) sulfate is formed. 'Anhydrous' simply means 'without water'.

$$\underset{\text{blue}}{CuSO_4 \cdot 5H_2O(s)} \rightarrow \underset{\text{white}}{CuSO_4(s)} + 5H_2O(l)$$

Heat is needed for this reaction to occur

If you add water to the white solid, it turns blue again; it also becomes very warm. See Figure 18.3 (page 192).

$$\underset{\text{white}}{CuSO_4(s)} + 5H_2O(l) \rightarrow \underset{\text{blue}}{CuSO_4 \cdot 5H_2O(s)}$$

Heat is released from this reaction

The original change has been exactly reversed. Even the heat that you put in originally has been given out again. This is called a **reversible** reaction and is indicated by a special arrow:

$$CuSO_4 \cdot 5H_2O(s) \rightleftharpoons CuSO_4(s) + 5H_2O(l)$$

HEATING AMMONIUM CHLORIDE

If you heat ammonium chloride, the white crystals disappear from the bottom of the tube and reappear further up. Heating ammonium chloride splits it into the colourless gases ammonia and hydrogen chloride.

KEY POINT

This reaction is an example of *thermal decomposition*.

KEY POINT

This is a reaction between an acid (HCl) and a base (NH_3) (see Chapter 17, page 186).

HINT

If you add water to the white solid, it turns blue again; it also becomes very warm. See Figure 11.11 (page 93).

$$\underset{\text{white solid}}{NH_4Cl(s)} \rightarrow \underset{\text{colourless gases}}{NH_3(g) + HCl(g)}$$ Heat is needed for this reaction to occur

These gases recombine further up the tube, where it is cooler, to form a white solid:

$$\underset{\text{colourless gases}}{NH_3(g) + HCl(g)} \rightarrow \underset{\text{white solid}}{NH_4Cl(s)}$$ Heat is released from this reaction

The reaction reverses when the conditions are changed from hot to cool.



◀ Figure 21.2 Heating ammonium chloride

This can, again, be shown with the reversible arrow:

$$NH_3(g) + HCl(g) \rightleftharpoons NH_4Cl(s)$$

CHEMISTRY ONLY

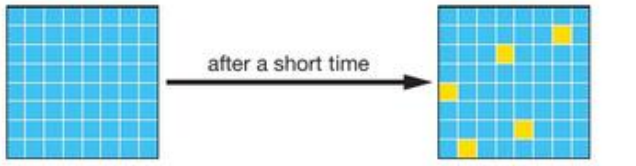
REVERSIBLE REACTIONS IN A SEALED CONTAINER

A *sealed container* means that no substances are added to the reaction mixture and no substances escape from it. On the other hand, heat may be either given off or absorbed.

Imagine a substance that can exist in two forms, one of which we'll represent by a blue square and the other by a yellow square. Suppose you start off with a sample which is entirely blue.

KEY POINT

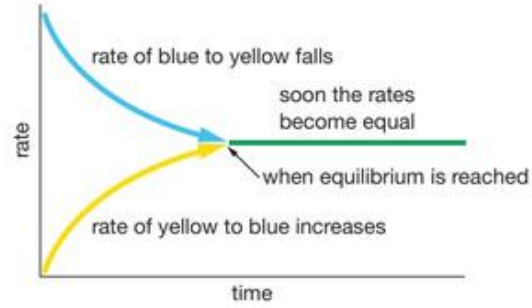
Imagine you started with 64 blue squares, and in any second there was a one in four chance of each of them changing colour. In the first second, 16 would change colour, leaving 48 blue squares. In the next second, a quarter of these change colour, but that's only 12; this leaves 36 blue ones. In the third second, nine would change colour, and so on. The rate of change falls as the number (concentration) of blue squares falls.



▲ Figure 21.3 Blue squares converting to yellow ones

Because you are starting with a high concentration of blue squares, at the beginning of the reaction the rate at which they turn yellow will be relatively high in terms of the number of squares changing colour per second. The number of blue squares changing colour per second (the rate of change) will fall as the blue gradually gets used up.

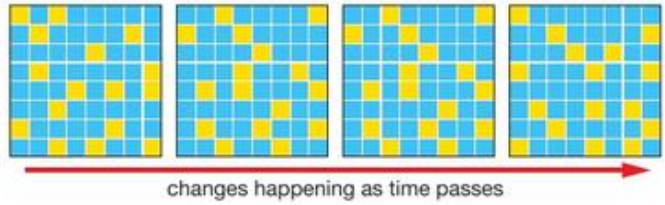
But the yellow squares can also change back to blue ones again. This is because it is a *reversible* reaction, *in which the products can react with each other and go back to form reactants*. At the start, there aren't any yellow squares, so the rate of change from yellow into blue is zero. As their number increases, the rate at which yellow change to blue also increases.



▲ Figure 21.4 The rates of the forward reaction and the reverse reaction become equal when equilibrium is reached.

Soon the rates of both reactions become equal. At that point, blue squares are changing into yellow ones at exactly the same rate that yellow ones are turning blue.

What would you see in the reaction mixture when that happens? The total numbers of blue squares and of yellow squares would remain constant, but the reaction would still be happening. If you followed the fate of any one particular square, sometimes it would be blue and sometimes yellow.



▲ Figure 21.5 The reaction continues, but total numbers of blue and yellow squares remain constant.

HINT

Be careful with the wording of your answer in the exam - the concentrations of the reactants and products at equilibrium are *constant*, they are not *equal*.

This is an example of a **dynamic equilibrium**. It is dynamic in the sense that the reactions are still continuing, but *the rate of the forward reaction is equal to the rate of the reverse reaction*. It is an equilibrium in the sense that the total amounts or *concentrations of the various things present (reactants and products) are now constant*.

Notice that you can set up a dynamic equilibrium only if the system is closed *in a sealed container*. If, for example, you removed the yellow squares as soon as they were formed, they would never get the chance to turn blue again. What was a reversible reaction would now go entirely in one direction as blue squares turn yellow without being replaced.

Consider the reversible reaction that occurs when calcium carbonate decomposes:

$$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$$

If this were not done in a closed container then all the CO_2 would escape and the reverse reaction would not occur.

Reversible reactions will eventually reach a state of equilibrium if they are left in a sealed container.

THE POSITION OF EQUILIBRIUM

Taking a general case where A reacts reversibly to give B:

$$A \rightleftharpoons 2B$$

The reaction from A to B (the left-to-right reaction) is described as the **forward reaction**. The reaction between from B to A (the right-to-left reaction) is called the **reverse reaction**.

If we let this reaction come to equilibrium then measure the amount of each substance present we might find that we have

$$\underset{90\%}{A} \rightleftharpoons \underset{10\%}{2B}$$

Because there is more A than B present at equilibrium, we say that the **position of equilibrium** lies to the left. It is important to realise that equilibrium does not mean 50% of reactants and products; the key thing is that the concentrations of A and B stay constant at equilibrium. These concentrations will remain constant until we change the conditions, for instance the temperature or the pressure. If we increase the temperature and allow the reaction time to reach equilibrium again, we might find that we have:

$$\underset{80\%}{A} \rightleftharpoons \underset{20\%}{2B}$$

Because the change we made decreased the concentration of what is on the left (A) and increased the concentration of what is on the right (B), we say that *the position of equilibrium has shifted to the right*. It does not matter that we still have more A than B, what we are looking at is how things change when we change the conditions.

HOW TO PREDICT THE EFFECT OF CHANGING CONDITIONS ON THE POSITION OF EQUILIBRIUM

This section looks at what can be done to change the position of equilibrium to produce as much as possible of what you want in the equilibrium mixture.

When the conditions in a system in dynamic equilibrium are changed, for example the temperature or the pressure, the position of equilibrium might shift left or right to give less or more of the products. A useful way to help you to decide what happens if various conditions are changed is to remember that *the reaction always sets about counteracting any changes you make*. This is similar to the idea that if a room is too warm you might take off your jacket so that you can cool down, or if the room is too cold you might put some more clothes on!

Things we might try to do to influence the reaction include:

- changing the pressure
- changing the temperature
- adding a catalyst (in fact, this turns out to have no effect on the position of equilibrium).

CHANGING THE PRESSURE

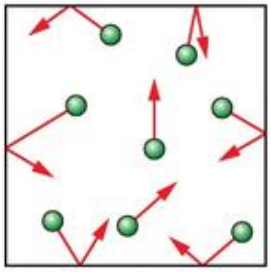
KEY POINT

The particles in solids and liquids are very close together and they are affected very little by changes in pressure. Their concentration does not really change as the pressure changes.

This only really applies to reactions in which at least one of the reactants or products is a *gas*, and where *the total numbers of gaseous molecules on both sides of the equation are different*. In the example here, there are three gaseous molecules on the left, but only two on the right.

$$A(g) + 2B(g) \rightleftharpoons C(g) + D(g)$$

Pressure is caused by molecules hitting the walls of their container. If you have fewer molecules in the same volume at the same temperature, you will have a lower pressure.



▲ Figure 21.6 Pressure is caused by molecules hitting the walls of their container – the fewer molecules, the lower the pressure.

If you increase the pressure, the reaction will respond by reducing it again. It can reduce the pressure by producing fewer gaseous molecules to hit the walls of the container, in this case by converting A + 2B (3 molecules) into C + D (2 molecules). Increasing the pressure will always cause the position of equilibrium to shift in the direction which produces the smaller number of gaseous molecules.

Consider the reaction between nitrogen gas and hydrogen gas to make ammonia:

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

If we want to make as much ammonia as possible, we need to make the pressure as high as possible. There are four molecules of gas on the left-hand side but only two on the right-hand side. Increasing the pressure will therefore shift the position of equilibrium to the right (the ammonia side), the side with fewer gas molecules.

If there is the same number of gaseous molecules on both sides of the equation, changing the pressure will make no difference to the position of equilibrium. For example, in this reaction:

$$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$$

increasing or decreasing the pressure has no effect on the position of equilibrium – it does not affect how much hydrogen, iodine and hydrogen iodide we have at equilibrium.

To summarise:

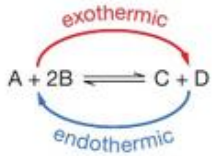
Increasing pressure: the position of equilibrium shifts to the side which has *fewer gas molecules.*

Decreasing pressure: the position of equilibrium shifts to the side which has *more gas molecules.*

CHANGING THE TEMPERATURE

REMINDER

Exothermic and endothermic reactions and ΔH values are explained in Chapter 19.



▲ Figure 21.7 If a reaction is exothermic in the forward reaction it will be endothermic in the reverse direction.

KEY POINT

At equilibrium, the rate of the forward reaction is equal to the rate of the reverse reaction so as much heat is being given out as is taken in – nothing appears to change.

KEY POINT

In an endothermic reaction the heat is converted to a different form (chemical energy), so it is being removed.

When we write a reversible reaction showing an enthalpy change, the ΔH always shows the enthalpy change for the *forward* reaction. The value of ΔH is given as if the reaction was a one-way process.

$$A + 2B \rightleftharpoons C + D \quad \Delta H = -100\,kJ/mol$$

So, in this case ΔH being negative tells us that the *forward* reaction is *exothermic*.

The reverse reaction will be *endothermic* by exactly the same amount.

Suppose you changed the conditions by decreasing the temperature of the equilibrium, for example if the reaction was originally in equilibrium at 500 °C you lower the temperature to 100 °C. The reaction will respond in a way that increases the temperature again. How can it do that?

If some A and B are converted into C and D, more heat is given out because this is an exothermic change. The extra heat that is produced will warm the reaction mixture up again. In other words, decreasing the temperature will cause the position of equilibrium to move to the right-hand side and more C and D to be formed.

	A + 2B ⇌ C + D	
Equilibrium at 500 °C	80%	20%
Equilibrium at 100 °C	60%	40%

Increasing the temperature will have exactly the opposite effect. The reaction equilibrium will change to remove the extra heat by absorbing it in an endothermic change. This time the reverse reaction is favoured and the position of equilibrium moves to the left.

To summarise:

Increasing temperature: the position of equilibrium shifts in the *endothermic direction*.

Decreasing temperature: the position of equilibrium shifts in the *exothermic direction*.

ADDING A CATALYST

Adding a catalyst speeds up reactions. In a reversible change, it speeds up the forward and reverse reactions by *the same proportion*. For example, if it speeds up the forward reaction ten times, it speeds up the reverse reaction ten times as well.

The net effect of this is that there is *no change* in the position of equilibrium if you add a catalyst. The catalyst is added to increase the rate at which equilibrium is reached.

AN EXAMPLE TO ILLUSTRATE HOW CHANGING REACTION CONDITIONS CAN AFFECT THE POSITION OF EQUILIBRIUM IN A REVERSIBLE REACTION

Nitrogen dioxide, NO_2, is a dark brown, poisonous gas. It can join together in pairs (**dimerise**) to make molecules of dinitrogen tetroxide, N_2O_4, which is colourless. There is a dynamic equilibrium between the two forms:

$$\underset{\text{brown}}{2NO_2(g)} \rightleftharpoons \underset{\text{colourless}}{N_2O_4(g)} \qquad \Delta H = -57\,\text{kJ/mol}$$

THE EFFECT OF PRESSURE

HINT

Warning! You have to be very careful here about predicting exactly what colour changes you would see. If you increase the pressure, for example, you squeeze the same number of molecules into a smaller space and so the colour will darken initially. Then it fades a bit as the equilibrium re-establishes, but not to its original colour. The gases are still compressed.

If you increase the pressure, the position of equilibrium will shift to reduce it again by producing fewer gaseous molecules. In other words, the position of equilibrium will move to the right and the reaction will produce more dinitrogen tetroxide.

If you lower the pressure, the position of equilibrium will shift to increase it again by producing more gaseous molecules. Therefore the position of equilibrium shifts to the left, and you will obtain a higher proportion of the brown nitrogen dioxide in the equilibrium mixture.

THE EFFECT OF TEMPERATURE

Notice from the equation that the change from nitrogen dioxide to dinitrogen tetroxide is exothermic. The negative sign for ΔH shows that heat is given out by the forward reaction.

If you decrease the temperature, the position of equilibrium will shift to produce more heat to counteract the change you have made. In other words, lowering the temperature causes the position of equilibrium to shift in the exothermic direction and there will be more dinitrogen tetroxide in the equilibrium mixture. The colour of the reaction mixture will fade.

If you increase the temperature, the position of equilibrium will shift to lower it again – the position of equilibrium shifts in the reverse, endothermic direction. In other words, more nitrogen dioxide will be formed and the colour of the gas will darken.



▲ Figure 21.8 Tubes containing the NO_2/N_2O_4 equilibrium at different temperatures.

You can see this happening in Figure 21.8. Of the three tubes containing this equilibrium mixture, one is at lab temperature, one is in ice and one is in hot water. Notice that the hot one is very dark brown and therefore contains a high proportion of nitrogen dioxide.

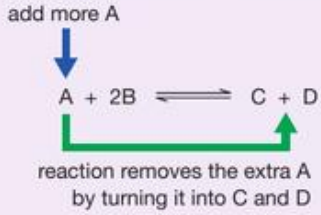
The one in the ice is slightly paler than the one in the air, showing that it must have a slightly greater proportion of the colourless dinitrogen tetroxide.

THE EFFECT OF CONCENTRATION

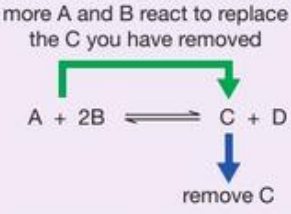
EXTENSION WORK

Changing the concentrations of reactants or products in a reversible reaction also has an effect on the position of equilibrium.

Take the reaction in Figure 21.9 as an example. If you add more A, the system responds by removing it again. That produces more C and D, which is what you probably want. You might choose to increase the amount of A if it was essential to convert as much B as possible into product because it was expensive, for example.



▲ Figure 21.9 Adding more of substance A



▲ Figure 21.10 Removing substance C as soon as it is formed

Alternatively, if you remove C as soon as it is formed, the reaction will respond by replacing it again by reacting more A and B (Figure 21.10). Removing a substance as soon as it is formed is a useful way of moving the position of equilibrium to generate more product.

END OF CHEMISTRY ONLY

CHAPTER QUESTIONS

SKILLS REASONING

1 The thermal decomposition of ammonium chloride is a reversible reaction.

$$NH_4Cl(s) \rightleftharpoons NH_3(g) + HCl(g)$$

SKILLS CRITICAL THINKING

a Explain how the equation shows that the reaction is reversible.

b Give one observation associated with the forward reaction.

c Name all the substances present in the reaction mixture if an equilibrium state is reached.

CHEMISTRY ONLY

SKILLS CRITICAL THINKING

8

2 For the following reactions at equilibrium, predict in which direction (**Left** or **Right** or **No change**) the position of equilibrium might shift if the conditions were changed:

		Change in conditions
$N_2O_4(g) \rightleftharpoons 2NO_2(g)$	Endothermic	Increase in temperature
$2NH_3(g) \rightleftharpoons N_2(g) + 3H_2(g)$	Endothermic	Increase in pressure
$N_2(g) + O_2(g) \rightleftharpoons 2NO(g)$	Endothermic	Decrease in temperature
$H_2(g) + I_2(g) \rightleftharpoons 2HI(g)$	Endothermic	Decrease in pressure
$CO(g) + 2H_2(g) \rightleftharpoons CH_3OH(g)$	Exothermic	Adding a catalyst

7

3 An important stage in the Contact process for the production of sulfuric acid involves a reversible reaction:

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g) \quad \Delta H = -196\,kJ/mol$$

At 450 °C the reaction mixture consists of a *dynamic equilibrium* involving sulfur dioxide, oxygen and sulfur trioxde.

a Explain what is meant by the term *dynamic equilibrium*. Be sure that you have explained what both of the words mean.

SKILLS REASONING

8

b Predict and explain the effect of an increase in pressure on the proportion of the sulfur trioxide present in the equilibrium mixture.

c Predict and explain the effect of lowering the temperature of the mixture on the position of equilibrium of the reaction.

4 Hydrogen can be made by the reaction between methane (natural gas) and steam. The reaction can be carried out by passing a mixture of methane and steam over a nickel catalyst at pressures between 2 and 30 atmospheres and a temperature of about 1000 °C.

$$CH_4(g) + H_2O(g) \rightleftharpoons CO(g) + 3H_2(g) \quad \Delta H = +210\,kJ/mol$$

a The pressure used is relatively low. What would be the effect on the conversion of the methane into carbon monoxide and hydrogen if the pressure was higher?

b Explain why a high temperature is used to get a good conversion of methane into hydrogen.

c Explain the use of the nickel catalyst in the production of hydrogen, referring to both the rate of the reaction and the position of equilibrium.

5 Ammonia, NH_3, is manufactured by passing a mixture of nitrogen and hydrogen over an iron catalyst at a pressure of 200 atmospheres or more, and a temperature of 450 °C.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) \quad \Delta H = -92\,kJ/mol$$

a Explain why this reaction will produce a higher percentage conversion into ammonia if the pressure is very high.

b 200 atmospheres is a high pressure, but not very high. Give a reason why most ammonia manufacturers don't use a pressure of, for example, 1000 atmospheres. (Hint: think about the risk of using high pressure.)

c Predict and explain whether you would get the best yield of ammonia in the equilibrium mixture at a low or a high temperature.

d The temperature used, 450 °C, is neither very high nor very low. Suggest a reason why a manufacturer might choose a temperature which gave less than an ideal percentage conversion. (Hint: think about rates of reaction.)

END OF CHEMISTRY ONLY